

The following depict ARP TLS TCP capture of network traffic for forensic analysis:

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arp

No.	Time	Source	Destination	Protocol	Length	Info
106	2.188940	46:b1:be:42:f3:f3	Broadcast	ARP	42	Gratuitous ARP for 10.125.22.31 (Reply)
392	4.033391	76:af:8c:06:91:1a	Broadcast	ARP	56	Gratuitous ARP for 10.125.29.219 (Reply)
434	4.848769	6c:94:66:f8:43:87	Broadcast	ARP	56	Who has 10.110.11.242? Tell 0.0.0.0
519	5.464178	ac:12:03:3e:99:82	Broadcast	ARP	56	Who has 10.125.67.70? Tell 0.0.0.0
525	5.568960	2e:7d:79:91:30:f7	Broadcast	ARP	56	Who has 10.125.29.138? Tell 0.0.0.0
542	5.771288	6c:94:66:f8:43:87	Broadcast	ARP	56	Who has 10.110.11.242? Tell 0.0.0.0
576	5.974973	2e:7d:79:91:30:f7	Broadcast	ARP	56	Who has 10.125.29.138? Tell 0.0.0.0
608	6.282137	2e:7d:79:91:30:f7	Broadcast	ARP	56	Who has 10.125.29.138? Tell 0.0.0.0
625	6.487458	ac:12:03:3e:99:82	Broadcast	ARP	56	Who has 10.125.67.70? Tell 0.0.0.0
649	6.794510	6c:94:66:f8:43:87	Broadcast	ARP	56	Who has 10.110.11.242? Tell 0.0.0.0
710	7.515168	ac:12:03:3e:99:82	Broadcast	ARP	56	Who has 10.125.67.70? Tell 0.0.0.0
749	8.535047	12:c3:5e:b0:88:1a	Broadcast	ARP	56	Gratuitous ARP for 10.125.19.28 (Reply)
769	8.744042	1e:9d:03:b0:1d:5e	Broadcast	ARP	56	Who has 10.125.28.121? Tell 0.0.0.0
797	9.149505	1e:9d:03:b0:1d:5e	Broadcast	ARP	56	Who has 10.125.28.121? Tell 0.0.0.0
825	9.460247	1e:9d:03:b0:1d:5e	Broadcast	ARP	56	Who has 10.125.28.121? Tell 0.0.0.0
841	9.766300	da:48:86:82:4a:4f	Broadcast	ARP	56	Who has 10.125.29.248? Tell 0.0.0.0
863	10.173491	da:48:86:82:4a:4f	Broadcast	ARP	56	Who has 10.125.29.248? Tell 0.0.0.0
881	10.381404	d0:65:78:80:99:59	Broadcast	ARP	56	Who has 10.125.23.64? Tell 0.0.0.0
889	10.480594	da:48:86:82:4a:4f	Broadcast	ARP	56	Who has 10.125.29.248? Tell 0.0.0.0
959	11.402857	d0:65:78:80:99:59	Broadcast	ARP	56	Who has 10.125.23.64? Tell 0.0.0.0
1048	12.426149	d0:65:78:80:99:59	Broadcast	ARP	56	Who has 10.125.23.64? Tell 0.0.0.0
1168	14.167834	4c:5f:70:a1:e4:60	Broadcast	ARP	56	Who has 10.125.19.91? Tell 0.0.0.0
1205	14.781503	7a:60:77:6b:26:c6	Broadcast	ARP	56	Who has 10.125.64.24? Tell 0.0.0.0
1213	14.886909	4a:08:2d:e7:fd:9f	Broadcast	ARP	56	Gratuitous ARP for 10.110.72.19 (Reply)
1233	15.092802	4c:5f:70:a1:e4:60	Broadcast	ARP	56	Who has 10.125.19.91? Tell 0.0.0.0
1306	15.808508	7a:60:77:6b:26:c6	Broadcast	ARP	56	Who has 10.125.64.24? Tell 0.0.0.0
1339	16.116844	4c:5f:70:a1:e4:60	Broadcast	ARP	56	Who has 10.125.19.91? Tell 0.0.0.0
1388	16.829473	7a:60:77:6b:26:c6	Broadcast	ARP	56	Who has 10.125.64.24? Tell 0.0.0.0
1389	16.832483	50:5a:65:ea:33:23	Broadcast	ARP	56	Who has 10.125.29.246? Tell 0.0.0.0
1454	17.853443	50:5a:65:ea:33:23	Broadcast	ARP	56	Who has 10.125.29.246? Tell 0.0.0.0
1537	18.881327	50:5a:65:ea:33:23	Broadcast	ARP	56	Who has 10.125.29.246? Tell 0.0.0.0

Frame 106: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface 0
Interface: 0 (\Device\NPF_{51579FFB-28B2-401B-B781-28C5F705E03})
Encapsulation type: Ethernet (1)
Arrival Time: Mar 15, 2025 14:10:48.296717000 India Standard Time
[Time shift for this packet: 0.000000000 seconds]
Epoch Time: 1742028048.296717000 seconds
[Time delta from previous captured frame: 0.095667000 seconds]
[Time delta from previous displayed frame: 0.000000000 seconds]
[Time since reference or first frame: 2.188940000 seconds]
Frame Number: 106
Frame Length: 42 bytes (336 bits)
Capture Length: 42 bytes (336 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:arp]
[Coloring Rule Name: ARP]
[Coloring Rule String: arp]

Ethernet II, Src: 46:b1:be:42:f3:f3 (46:b1:be:42:f3:f3), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
> Destination: Broadcast (ff:ff:ff:ff:ff:ff)
> Source: 46:b1:be:42:f3:f3 (46:b1:be:42:f3:f3)
Type: ARP (0x0806)

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tcp.flags.syn

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	20.42.65.91	10.125.29.206	TLSv1.2	93	Application Data
4	0.047931	10.125.29.206	20.42.65.91	TCP	54	55219 → 443 [ACK] Seq=1 Ack=40 Win=256 Len=0
9	0.265901	20.42.65.91	10.125.29.206	TLSv1.2	168	Application Data
10	0.267227	10.125.29.206	20.42.65.91	TLSv1.2	89	Application Data
16	0.380499	52.108.44.3	10.125.29.206	TLSv1.2	98	Application Data
17	0.381125	10.125.29.206	52.108.44.3	TLSv1.2	89	Application Data
18	0.381215	10.125.29.206	52.108.44.3	TLSv1.2	89	Application Data
19	0.389643	52.108.44.3	10.125.29.206	TCP	60	443 → 55214 [ACK] Seq=45 Ack=71 Win=16384 Len=0
26	0.552187	10.125.29.206	52.108.8.12	TLSv1.2	373	Application Data
27	0.552314	10.125.29.206	52.108.8.12	TLSv1.2	3378	Application Data
28	0.562244	20.42.65.91	10.125.29.206	TCP	60	443 → 55219 [ACK] Seq=154 Ack=36 Win=16384 Len=0
29	0.568561	52.108.8.12	10.125.29.206	TCP	60	443 → 55204 [ACK] Seq=1 Ack=3644 Win=16385 Len=0
30	0.631224	52.108.8.12	10.125.29.206	TLSv1.2	329	Application Data
35	0.673157	10.125.29.206	52.108.8.12	TCP	54	55204 → 443 [ACK] Seq=3644 Ack=276 Win=1033 Len=0
46	0.863659	10.125.29.206	184.27.196.32	TLSv1.2	161	Application Data
47	0.874293	184.27.196.32	10.125.29.206	TLSv1.2	833	Application Data
48	0.874388	184.27.196.32	10.125.29.206	TLSv1.2	85	Application Data
49	0.874418	10.125.29.206	184.27.196.32	TCP	54	55203 → 443 [ACK] Seq=108 Ack=811 Win=4105 Len=0
50	0.874834	10.125.29.206	184.27.196.32	TLSv1.2	89	Application Data
51	0.923878	184.27.196.32	10.125.29.206	TCP	60	443 → 55203 [ACK] Seq=811 Ack=143 Win=501 Len=0
67	1.425731	10.125.29.206	52.123.160.120	TLSv1.2	103	Application Data
76	1.590085	52.123.160.120	10.125.29.206	TLSv1.2	92	Application Data
77	1.643420	10.125.29.206	52.123.160.120	TCP	54	55187 → 443 [ACK] Seq=50 Ack=39 Win=515 Len=0
95	1.923634	10.125.29.206	52.123.164.120	TLSv1.2	103	Application Data
96	1.931191	52.123.164.120	10.125.29.206	TLSv1.2	92	Application Data
99	1.985008	10.125.29.206	52.123.164.120	TCP	54	54901 → 443 [ACK] Seq=50 Ack=39 Win=513 Len=0
114	2.285468	10.125.29.206	52.113.194.132	SSL	55	Continuation Data
115	2.285470	10.125.29.206	52.113.194.133	SSL	55	Continuation Data
130	2.317990	52.113.194.133	10.125.29.206	TCP	66	443 → 55170 [ACK] Seq=1 Ack=2 Win=16384 Len=0 SLE=1 SRE=2
131	2.318115	52.113.194.132	10.125.29.206	TCP	66	443 → 55171 [ACK] Seq=1 Ack=2 Win=16384 Len=0 SLE=1 SRE=2
178	3.037681	104.97.76.241	10.125.29.206	TLSv1.2	350	Application Data

▼ Frame 99: 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface 0
 Interface id: 0 (\Device\NPF_{51579FFB-28B2-401B-B781-28CF5F705E03})
 Encapsulation type: Ethernet (1)
 Arrival Time: Mar 15, 2025 14:10:48.092785000 India Standard Time
 [Time shift for this packet: 0.000000000 seconds]
 Epoch Time: 1742028048.092785000 seconds
 [Time delta from previous captured frame: 0.002027000 seconds]
 [Time delta from previous displayed frame: 0.053817000 seconds]
 [Time since reference or first frame: 1.985008000 seconds]
 Frame Number: 99
 Frame Length: 54 bytes (432 bits)
 Capture Length: 54 bytes (432 bits)
 [Frame is marked: False]
 [Frame is ignored: False]
 [Protocols in frame: eth:ethertype:ip:tcp]
 [Coloring Rule Name: TCP]
 [Coloring Rule String: tcp]

▼ Ethernet II, Src: ae:b4:9c:fb:70:88 (ae:b4:9c:fb:70:88), Dst: 24:2a:04:fd:46:9f (24:2a:04:fd:46:9f)
 Destination: 24:2a:04:fd:46:9f (24:2a:04:fd:46:9f)
 Address: 24:2a:04:fd:46:9f (24:2a:04:fd:46:9f)
 = LG bit: Globally unique address (factory default)

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tcp.flags.syn==1

No.	Time	Source	Destination	Protocol	Length	Info
191	3.190122	10.125.29.206	10.110.14.45	TCP	66	55230 → 7680 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
204	3.227463	10.125.29.206	184.27.196.59	TCP	66	55238 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
206	3.240895	184.27.196.59	10.125.29.206	TCP	66	80 → 55238 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM=1 WS=128
209	3.244193	184.27.196.59	10.125.29.206	TCP	66	[TCP Out-Of-Order] 80 → 55238 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM=1 WS=128
226	3.254121	10.125.29.206	184.27.196.59	TCP	66	55239 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
227	3.260360	184.27.196.59	10.125.29.206	TCP	66	80 → 55239 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM=1 WS=128
241	3.276898	10.125.29.206	184.27.196.59	TCP	66	55240 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
242	3.283255	184.27.196.59	10.125.29.206	TCP	66	80 → 55240 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM=1 WS=128
258	3.297288	10.125.29.206	184.27.196.59	TCP	66	55241 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
269	3.334357	184.27.196.59	10.125.29.206	TCP	66	80 → 55241 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM=1 WS=128
271	3.334569	184.27.196.59	10.125.29.206	TCP	66	[TCP Out-Of-Order] 80 → 55241 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM=1 WS=128
290	3.421981	10.125.29.206	184.27.196.59	TCP	66	55242 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
291	3.429377	184.27.196.59	10.125.29.206	TCP	66	80 → 55242 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM=1 WS=128
294	3.435314	184.27.196.59	10.125.29.206	TCP	66	[TCP Out-Of-Order] 80 → 55242 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM=1 WS=128
312	3.450424	10.125.29.206	184.27.196.59	TCP	66	55243 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
313	3.455436	184.27.196.59	10.125.29.206	TCP	66	80 → 55243 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM=1 WS=128
327	3.470351	10.125.29.206	184.27.196.59	TCP	66	55244 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
328	3.475771	184.27.196.59	10.125.29.206	TCP	66	80 → 55244 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM=1 WS=128
344	3.485319	10.125.29.206	184.27.196.59	TCP	66	55245 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
345	3.491545	184.27.196.59	10.125.29.206	TCP	66	80 → 55245 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM=1 WS=128
452	4.981015	10.125.29.206	184.25.109.168	TCP	66	55248 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
453	4.987905	184.25.109.168	10.125.29.206	TCP	66	80 → 55248 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM=1 WS=128
467	5.000285	10.125.29.206	184.25.109.168	TCP	66	55249 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
468	5.006231	184.25.109.168	10.125.29.206	TCP	66	80 → 55249 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM=1 WS=128
607	6.253794	10.110.138.3	10.125.29.206	ICMP	94	Destination unreachable (Host unreachable)
1366	16.550781	10.125.29.206	23.193.165.65	TCP	66	55251 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
1367	16.556637	23.193.165.65	10.125.29.206	TCP	66	80 → 55251 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM=1 WS=128
2241	28.182165	10.125.29.206	10.100.61.190	TCP	66	55252 → 7680 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
2316	29.192845	10.125.29.206	10.100.61.190	TCP	66	[TCP Retransmission] 55252 → 7680 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
2549	31.206442	10.125.29.206	10.100.61.190	TCP	66	[TCP Retransmission] 55252 → 7680 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
2550	31.224649	10.125.29.206	184.27.196.129	TCP	66	55253 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=1

Frame 191: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface 0
Interface id: 0 (DeviceNPF {51579FFB-2B02-401B-8781-28CFS705FE03})
Encapsulation type: Ethernet (1)
Arrival Time: Mar 15, 2025 14:10:49.297899000 India Standard Time
[Time shift for this packet: 0.000000000 seconds]
Epoch Time: 1742028049.297899000 seconds
[Time delta from previous captured frame: 0.000127000 seconds]
[Time delta from previous displayed frame: 0.000000000 seconds]
[Time since reference or first frame: 3.190122000 seconds]
Frame Number: 191
Frame Length: 66 bytes (528 bits)
Capture Length: 66 bytes (528 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: ethertype:ip:tcp]
[Coloring Rule Name: TCP SYN/FIN]
[Coloring Rule String: tcp.flags & 0x02 || tcp.flags.fin == 1]

Ethernet II, Src: ae:64:9c:fb:70:88 (ae:64:9c:fb:70:88), Dst: 24:2a:04:fd:46:9f (24:2a:04:fd:46:9f)

Destination: 24:2a:04:fd:46:9f (24:2a:04:fd:46:9f)
Address: 24:2a:04:fd:46:9f (24:2a:04:fd:46:9f)
...0...

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tcp.flags.fin==1

No.	Time	Source	Destination	Protocol	Length	Info
214	3.245942	10.125.29.206	184.27.196.59	TCP	54	55238 → 80 [FIN, ACK] Seq=457 Ack=2921 Win=131328 Len=0
221	3.246269	184.27.196.59	10.125.29.206	TCP	54	80 → 55238 [FIN, ACK] Seq=7118 Ack=457 Win=65664 Len=0
237	3.265674	184.27.196.59	10.125.29.206	TCP	54	80 → 55239 [FIN, ACK] Seq=7118 Ack=481 Win=65664 Len=0
240	3.265841	10.125.29.206	184.27.196.59	TCP	54	55239 → 80 [FIN, ACK] Seq=481 Ack=7119 Win=130048 Len=0
247	3.287507	10.125.29.206	184.27.196.59	TCP	54	55240 → 80 [FIN, ACK] Seq=457 Ack=2921 Win=131328 Len=0
255	3.287672	184.27.196.59	10.125.29.206	TCP	54	80 → 55240 [FIN, ACK] Seq=7118 Ack=457 Win=65664 Len=0
282	3.342889	184.27.196.59	10.125.29.206	TCP	54	80 → 55241 [FIN, ACK] Seq=7118 Ack=481 Win=65664 Len=0
284	3.343163	10.125.29.206	184.27.196.59	TCP	54	55241 → 80 [FIN, ACK] Seq=481 Ack=7119 Win=130048 Len=0
298	3.438527	10.125.29.206	184.27.196.59	TCP	54	55242 → 80 [FIN, ACK] Seq=438 Ack=2921 Win=131328 Len=0
306	3.439000	184.27.196.59	10.125.29.206	TCP	54	80 → 55242 [FIN, ACK] Seq=7118 Ack=438 Win=65664 Len=0
323	3.459537	184.27.196.59	10.125.29.206	TCP	54	80 → 55243 [FIN, ACK] Seq=7118 Ack=471 Win=65664 Len=0
325	3.459717	10.125.29.206	184.27.196.59	TCP	54	55243 → 80 [FIN, ACK] Seq=471 Ack=7119 Win=130048 Len=0
335	3.479965	10.125.29.206	184.27.196.59	TCP	54	55244 → 80 [FIN, ACK] Seq=447 Ack=2921 Win=131328 Len=0
340	3.480194	184.27.196.59	10.125.29.206	TCP	54	80 → 55244 [FIN, ACK] Seq=7118 Ack=447 Win=65664 Len=0
355	3.496449	184.27.196.59	10.125.29.206	TCP	54	80 → 55245 [FIN, ACK] Seq=7118 Ack=471 Win=65664 Len=0
357	3.496695	10.125.29.206	184.27.196.59	TCP	54	55245 → 80 [FIN, ACK] Seq=471 Ack=7119 Win=130048 Len=0
366	3.580368	10.125.29.206	184.27.196.59	TCP	54	[TCP Spurious Retransmission] 55239 → 80 [FIN, ACK] Seq=481 Ack=7119 Win=130048 Len=0
399	4.191145	10.125.29.206	184.27.196.59	TCP	54	[TCP Spurious Retransmission] 55239 → 80 [FIN, ACK] Seq=481 Ack=7119 Win=130048 Len=0
465	4.992161	184.25.109.168	10.125.29.206	TCP	54	80 → 55248 [FIN, ACK] Seq=7118 Ack=203 Win=65664 Len=0
480	5.011035	184.25.109.168	10.125.29.206	TCP	54	80 → 55249 [FIN, ACK] Seq=7118 Ack=197 Win=65664 Len=0
517	5.391825	10.125.29.206	184.27.196.59	TCP	54	[TCP Spurious Retransmission] 55239 → 80 [FIN, ACK] Seq=481 Ack=7119 Win=130048 Len=0
721	7.800089	10.125.29.206	184.27.196.59	TCP	54	[TCP Spurious Retransmission] 55239 → 80 [FIN, ACK] Seq=481 Ack=7119 Win=130048 Len=0
1075	12.613972	10.125.29.206	184.27.196.59	TCP	54	[TCP Spurious Retransmission] 55239 → 80 [FIN, ACK] Seq=481 Ack=7119 Win=130048 Len=0
1372	16.564423	23.193.165.65	10.125.29.206	TCP	54	80 → 55251 [FIN, ACK] Seq=188 Ack=112 Win=64256 Len=0
1374	16.564505	10.125.29.206	23.193.165.65	TCP	54	55251 → 80 [FIN, ACK] Seq=112 Ack=189 Win=131072 Len=0
2725	33.733206	34.102.128.190	10.125.29.206	TCP	54	443 → 55149 [FIN, ACK] Seq=32 Ack=1 Win=1008 Len=0
3213	40.127041	184.25.109.8	10.125.29.206	TCP	54	80 → 55257 [FIN, ACK] Seq=7118 Ack=203 Win=65664 Len=0
3230	40.163569	184.25.109.8	10.125.29.206	TCP	54	80 → 55258 [FIN, ACK] Seq=7118 Ack=197 Win=65664 Len=0
3493	43.284233	184.25.109.8	10.125.29.206	TCP	54	80 → 55259 [FIN, ACK] Seq=7118 Ack=203 Win=32768 Len=0
3593	44.481274	184.25.109.8	10.125.29.206	TCP	54	80 → 55260 [FIN, ACK] Seq=7118 Ack=197 Win=65664 Len=0
3735	45.757015	10.125.29.206	184.27.196.59	TCP	54	55261 → 80 [FIN, ACK] Seq=447 Ack=2921 Win=131328 Len=0

▼ Frame 214: 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface 0
Interface id: 0 (Device\NPF_{51579FFB-28B2-4018-B781-28CF5F705E03})
Encapsulation type: Ethernet (1)
Arrival Time: Mar 15, 2025 14:10:49.353719000 India Standard Time
[Time shift for this packet: 0.00000000 seconds]
Epoch Time: 1742028049.353719000 seconds
[Time delta from previous captured frame: 0.000015000 seconds]
[Time delta from previous displayed frame: 0.000000000 seconds]
[Time since reference or first frame: 3.245942000 seconds]
Frame Number: 214
Frame Length: 54 bytes (432 bits)
Capture Length: 54 bytes (432 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:tcp]
[Coloring Rule Name: HTTP]
[Coloring Rule String: http || tcp.port == 80 || http2]

▼ Ethernet II, Src: ae:b4:9c:fb:70:88 (ae:b4:9c:fb:70:88), Dst: 24:2a:04:fd:46:9f (24:2a:04:fd:46:9f)
▼ Destination: 24:2a:04:fd:46:9f (24:2a:04:fd:46:9f)
Address: 24:2a:04:fd:46:9f (24:2a:04:fd:46:9f)
.....0. LG bit: Globally unique address (factory default)

Wireshark - Protocol Hierarchy Statistics - nmims.pcapng

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDU/s
▼ Frame	100.0	401	100.0	42028	20 k	0	0	0	401
▼ Ethernet	100.0	401	15.5	6523	3,121	0	0	0	401
Logical-Link Control	2.0	8	1.0	400	191	0	0	0	8
Spanning Tree Protocol	2.0	8	0.8	336	160	8	336	160	8
Link Layer Discovery Protocol	0.2	1	0.7	282	134	1	282	134	1
▼ Internet Protocol Version 6	29.9	120	11.4	4800	2,296	0	0	0	120
User Datagram Protocol	14.0	56	1.1	448	214	0	0	0	56
Multicast Domain Name System	6.0	24	2.2	912	436	24	912	436	24
Link-local Multicast Name Resolution	2.2	9	0.5	217	103	9	217	103	9
DHCPv6	5.7	23	5.8	2434	1,164	23	2434	1,164	23
Internet Control Message Protocol v6	16.0	64	4.5	1900	909	64	1900	909	64
▼ Internet Protocol Version 4	56.9	228	11.0	4640	2,220	0	0	0	228
User Datagram Protocol	47.4	190	3.6	1520	727	0	0	0	190
Simple Service Discovery Protocol	8.7	35	12.1	5095	2,438	35	5095	2,438	35
NetBIOS Name Service	13.0	52	8.2	3446	1,649	52	3446	1,649	52
▼ NetBIOS Datagram Service	1.5	6	2.9	1236	591	0	0	0	6
SMB (Server Message Block Protocol)	1.5	6	1.8	744	356	0	0	0	6
SMB MailSlot Protocol	1.5	6	0.4	150	71	0	0	0	6
Microsoft Windows Browser Protocol	1.5	6	0.5	228	109	6	228	109	6
Multicast Domain Name System	15.7	63	6.3	2667	1,276	63	2667	1,276	63
Link-local Multicast Name Resolution	3.7	15	0.9	379	181	15	379	181	15
Dynamic Host Configuration Protocol	0.2	1	0.7	300	143	1	300	143	1
Domain Name System	4.5	18	4.6	1920	918	18	1920	918	18
▼ Transmission Control Protocol	4.5	18	2.7	1144	547	14	328	156	18
Hypertext Transfer Protocol	1.0	4	1.8	736	352	4	736	352	4
Internet Group Management Protocol	5.0	20	0.9	376	179	20	376	179	20
Address Resolution Protocol	11.0	44	4.8	2024	968	44	2024	968	44

No display filter.

Help

Protocols Copy X Close

ARP Poisoning Analysis

Signs of ARP Poisoning in Captured Traffic:

- **Multiple ARP Responses with Different MAC Addresses for the Same IP:** In a typical ARP poisoning attack, the attacker sends fake ARP responses to the victim's machine, associating the attacker's MAC address with the IP address of a legitimate machine (e.g., a gateway). This results in conflicting ARP entries, where the same IP address is mapped to multiple MAC addresses.
- **Suspicious ARP Replies:** If you see multiple ARP replies for the same IP address but with different MAC addresses, this could be an indication of ARP poisoning.

Distinguishing Legitimate ARP Requests from Malicious Ones:

- **Legitimate ARP Requests:** Legitimate ARP requests are typically broadcasted when a device does not know the MAC address of a particular IP address, and the response from the target device will come back with the correct mapping.
- **Malicious ARP Requests/Responses:** In an ARP poisoning attack, attackers may send unsolicited ARP responses or ARP requests with spoofed IP-MAC pairings. These responses might not match the actual network topology or may arrive unsolicited, potentially causing network issues.

MAC Flooding Analysis

How MAC Flooding Affects Network Performance:

- **Overloading the Switch's MAC Table:** In a MAC flooding attack, the attacker floods the switch with frames containing fake source MAC addresses. The switch then attempts to map each MAC address to a specific port in its MAC address table. When the table is full, the switch enters a fail-safe mode and begins broadcasting traffic to all ports. This leads to increased network traffic, degradation of performance, and potential disruption of communication between devices.
- **Loss of Network Segmentation:** If a switch's MAC table is flooded, it can no longer perform its intended task of routing frames only to the correct destination. Instead, all traffic is broadcast to every port, affecting network efficiency.

Indicators of MAC Flooding Attack in Wireshark:

- **High Volume of Frames with Different Source MAC Addresses:** Look for packets that have many different source MAC addresses but are directed to the same destination. This indicates that the attacker is attempting to flood the switch's MAC table.
- **Corrupted Packets:** Using the "Analyze Expert Information" tab in Wireshark, you may see issues like "corrupted packets," which are indicative of abnormal network traffic, such as that generated by a MAC flood.

SYN-FIN Flood DoS Attack Analysis

How SYN-FIN Flood Attack Impacts Server Resources:

- **Server's TCP Stack is Overloaded:** The SYN-FIN flood attack sends TCP packets with both the SYN (synchronize) and FIN (finish) flags set, which is an invalid combination. Since the server expects only one flag at a time (SYN for starting a connection or FIN for ending one), this forces the server's TCP stack to process these invalid packets. The server may run out of resources trying to handle these requests.
- **Depletion of System Resources:** The server must allocate resources to manage these "half-open" connections, which leads to resource exhaustion. Over time, this can prevent legitimate connections from being established or processed, resulting in a Denial of Service (DoS).

Key Indicators of SYN-FIN Flood Attack in Wireshark:

- **Packets with Both SYN and FIN Flags Set:** Use the filter `tcp.flags.syn == 1 and tcp.flags.fin == 1` to specifically look for SYN-FIN packets. These should not appear in regular, legitimate traffic, as they are an invalid combination.
- **High Volume of SYN-FIN Packets:** Observe the volume of such packets. A spike in traffic with these flags indicates an attack.
- **Lack of Responses from the Server:** The server will likely ignore or fail to respond to these invalid packets, leading to one-sided communication where the attacker sends packets, but the server does not reply.
- **Traffic Visualization with I/O Graph:** By visualizing the traffic in the I/O Graph, you can see a noticeable spike or high volume of SYN-FIN packets that correlate with the flood.